

DHEERAJ SONGARA

Graduate Student | Int. Ph.D. in Neuroscience

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RESEARCH INTEREST *To explore the neural basis of memory formation and engram theory through an integrative approach combining electrophysiology, in vivo imaging, transcriptomics, and behavioral neuroscience.*

WORK EXPERIENCE

Research Scholar · *National Centre for Biological Sciences, Bengaluru* *Sep 2021 – Present*

- ▶ Investigating pyramidal neuron subtypes in the adult mouse hippocampus.
- ▶ Combining ex vivo electrophysiology, single-cell RNA sequencing, and behavioural assays to study neuronal subtypes.
- ▶ Using chemogenetics and viral injections for targeted neuronal manipulation.

Research Scholar · *National Centre for Biological Sciences, Bengaluru* *Dec 2018 – Sep 2021*

- ▶ Studied amygdalar dysfunction in Fragile X Syndrome and the role of astrocytic glutamate uptake in stress.
- ▶ Conducted patch-clamp recordings, auditory fear conditioning, and chronic stress paradigms in rodent models.

Industry · *CIPLA, Pithampur, India* *Jul 2014 – Dec 2014*

- ▶ Managed reference standards and control chemicals used for quality control assays in a GMP-compliant pharmaceutical unit

EDUCATION

Int. Ph.D. — Neuroscience · *National Centre for Biological Sciences, Bengaluru* *Jul 2018 – Present*

B.Sc. — Biotechnology · *Choithram College of Professional Studies, Indore* *Jul 2015 – 2018*

TECHNICAL SKILLS

Electrophysiology & Imaging

- ▶ Whole-cell patch clamp (brain slices & cultured neurons)
- ▶ In vitro calcium imaging
- ▶ Confocal & fluorescence microscopy
- ▶ Morphological analysis (Sholl analysis)

Molecular & Cellular Biology

- ▶ Immunofluorescence, western blotting, qPCR
- ▶ FACS & MACS for cell sorting
- ▶ Neuronal & HEK cell culture, viral production, transfection

Animal Handling & Behaviour

- ▶ Mouse & rat handling, breeding, genotyping (PCR, gel electrophoresis)
- ▶ Stereotaxic viral injections, perfusion, brain slicing
- ▶ Behavioural assays: OFT, EPM, Y-maze, NOR, NLR, fear conditioning

Genomics & Computational Analysis

- ▶ Single-cell RNA sequencing, bulk RNA sequencing
- ▶ Python — electrophysiology & single-cell analysis pipelines
- ▶ Software: ImageJ, Imaris, FlowJo, LasX Confocal, AnyMaze, FACSDiva, Neurolucida

PUBLICATIONS

1. **Songara D. & Ghosh H.** *Protocol for in-house AAV production and stereotaxic-targeting of CaMKII α -Cre positive as well as negative neurons in CA1 in mice. (In submission)*
2. **Songara D. & Ghosh H.** *Memory encoding and retrieval are executed by molecularly specialized subsets within the engram ensembles. (In submission)*
3. **Fernandes et al., 2021.** *Correction of Amygdalar Dysfunction in a Rat Model of Fragile X Syndrome.* [Cell Reports, 37, 109805.](#)

TEACHING & PRESENTATIONS

- ▶ Poster presentation scheduled at FENS Forum 2026, Spain.
- ▶ Oral presentation at NGN24 (No Garland Neuroscience), IISER TVM.
- ▶ Teaching Assistant – Neuroscience Course, NCBS (2 years).